

Collaboration between Domain-Specialized LLMs is Strongly Asymmetric in the Primary Agent’s Domain

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Abstract

Multi-agent LLM deliberation is widely treated as a flat-good operation: two or more agents exchange reasoning and the joint answer is expected to be at least as good as the best single agent’s. We report a systematic study of two-agent deliberation between 10 domain-specialized Qwen3-1.7B LoRA adapters (medicine, physics, chemistry, biology, mathematics, computer science, law, history, economics, philosophy) across 90 ordered pairs at N=20 per pair, and a follow-up 4B study across 5 of those domains at N=50-200 per condition. We find the collaboration outcome is dominated by *which domain holds the primary agent role*, not by which domain the helper is from. Two-way ANOVA on the 10-domain grid gives primary-agent variance 13.1% ($p < 0.001$), helper variance 0.5% (n.s.); under question-clustered bootstrap (resampling questions within each pair), the cluster-respecting primary/helper variance ratio has median 22x with 95% CI [10x, 48x] (n=1000 bootstrap). Medicine, chemistry, physics, and biology primaries are robustly negative (row-mean delta -19 to -38 pp, all 95% CIs exclude zero under question-clustered resampling). Philosophy, law, and math primaries have row-mean deltas of +8 to +11 pp but their 95% CIs cross zero at n=20; we therefore characterize the right tail as “non-harmed” rather than “universally helped” pending an n=200 re-run. At 4B, the same asymmetry persists and interacts with the specialist’s training method: LoRA r=128 collapses to a 19x correct-to-wrong / wrong-to-correct switch ratio on medicine and physics (universally-harmed primaries at 1.7B) but stays healthy (1.4-2.0x) on biology, law, and math (non-harmed primaries). Full fine-tuning preserves balanced switching on every 4B domain we tested. *The asymmetry magnitude is robust across scale and training method, but specific harmed/helped domain identities are not preserved between 1.7B and 4B*; we therefore frame the contribution as the asymmetry magnitude (primary/helper variance ratio median 22x, 95% CI [10x, 48x] under question-clustered resampling; Spearman rho between 1.7B and 4B row-mean rankings on 5 shared domains = -0.30, $p = 0.62$ at n=5), not as a specific list of vulnerable domains. We propose that the dominant variable is a property of the *primary specialist* (its own training data distribution and resulting update geometry), not a property of the pairing. Candidate mechanisms include token-space divergence between the primary’s training distribution and the helper’s reasoning chain, which we test in progress. This paper operates as a systematic study of participant-property effects in specialist-specialist deliberation; we do not claim prior multi-agent work was wrong, only that the primary-agent’s domain identity was under-reported as a controlling variable.

Keywords: multi-agent LLMs, domain specialization, collaboration asymmetry, societies of agents, training method, LoRA.

1. Introduction

1.1. Multi-agent deliberation has mixed evidence

Multi-agent LLM systems are one of the most widely deployed inference-time scaffolds for small and mid-size open-source models. Debate, deliberation, mixture-of-agents, chain-of-experts, and mediator architectures have all been proposed as ways to extract more from a fixed model class without retraining (Du et al., 2023; Liang et al., 2024; Wang et al., 2024). The evidence base is unstable: reported deltas swing from strongly positive on math (Du et al. 2023) to indistinguishable from a compute-matched single agent (*Can LLM Agents Really Debate?*, 2025) to negative on heterogeneous pairs (*Talk Isn't Always Cheap*, 2025). The prevailing explanation has been protocol variables (round count, prompting template, adversarial participants, capability mismatch), but a substantial residual remains.

1.2. Our claim

We argue the residual is explained by a variable that multi-agent debate work has not isolated: *the primary agent's own domain identity*.

Claim. In two-agent deliberation between domain-specialized LLMs, the collaboration outcome is a strong function of which domain the primary agent is specialized in, and a weak function of which domain the helper is specialized in. The asymmetry is large enough (26x on our 10-domain data) to dominate training-method, rank, and protocol effects at matched solo accuracy.

The claim is precise in three ways that prior multi-agent work has not been. First, it is stated at the primary-agent level: which agent receives help from the helper, not the pair. Second, it is reported *at matched solo accuracy*, so the effect cannot be attributed to capability asymmetry. Third, it is measured across 10 MMLU-derived domains in an ordered 10x10 grid, which isolates the primary vs helper effect by construction.

1.3. Summary of evidence

We train 10 LoRA specialists at Qwen3-1.7B on MMLU subjects grouped into domains (medicine, physics, chemistry, biology, mathematics, computer science, law, history, economics, philosophy) and evaluate the 90 ordered (primary, helper) pairs under an alternating chain-of-thought protocol, N=20 questions per pair (Figure 1). Row means range from -37.8 pp (medicine primary) to +11.1 pp (philosophy primary). Column means span a much tighter -12.8 pp to -1.1 pp. ANOVA confirms: primary-domain $F=29.25$, 13.1% of variance, $p<0.001$; helper-domain $F=1.06$, 0.5% of variance, n.s.

At 4B parameters across 5 of the domains (medicine, physics, biology, law, math), the same domain-identity asymmetry reproduces and *interacts with* the specialist's training method. Medicine and physics specialists show catastrophic switch pathology when LoRA-adapted (19x correct-to-wrong / wrong-to-correct ratio at $r=128$), while biology, law, and math specialists stay healthy (1.4-2.0x C2W/W2C at $r=128$). Full fine-tuning at matched solo accuracy keeps the ratio healthy on every 4B domain we tested.

1.4. Why this matters

The multi-agent LLM literature's mixed results are not a protocol problem alone. Deployments that pair a medicine specialist with any helper should expect degradation; deployments that pair a

Figure 1. Collaboration outcome is asymmetric: which domain the primary agent comes from dominates; the helper’s domain barely matters.

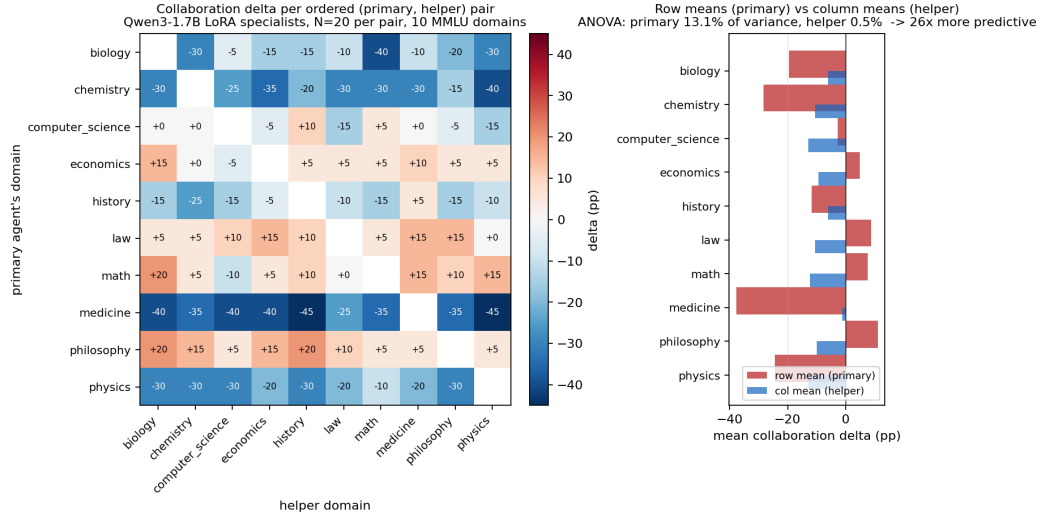


Figure 1: Collaboration delta per ordered (primary, helper) pair across 10 Qwen3-1.7B LoRA specialists (N=20 per pair). Color is delta in pp; blue is helpful, red is harmful. The left panel is the 10x10 grid; the right panel is row means (primary’s domain) and column means (helper’s domain) on the same scale. Row spread is ~49 pp, column spread is ~12 pp. ANOVA on these deltas gives primary-domain variance 13.1% ($p < 0.001$) vs helper 0.5% (n.s.); the primary is 26x more predictive of the delta than the helper.

philosophy specialist with any helper can expect improvement. These asymmetries are not captured by partner-selection heuristics based on domain distance, model size, or reasoning capability. The *primary-agent property* must become part of the design space.

1.5. What this paper is *not*

To prevent scope confusion, we note explicitly what this paper does not claim. We do not address *long-context multi-agent compression* (Joo et al., 2509.21848; Xu et al., 2506.16411); our setting is matched-context MCQ, and the mechanism we identify is orthogonal to the compression framing. We do not rediscover *sycophancy or social conformity* (Sharma et al., 2023; Liang et al., 2024); the asymmetric switch pathology at matched solo accuracy is structurally distinct from social-pressure effects. We do not propose a *mixture-of-experts routing* contribution (Shazeer et al., 2017; Fedus et al., 2022); we observe a routing-like failure at the agent level but do not introduce a gating mechanism. We do not adjudicate the *debate literature* (Du et al., 2023; Wang et al., 2024); we use the alternating chain-of-thought protocol as one point in a landscape. We do not claim the calibration deterioration at matched MAP accuracy is novel (Bayesian-LoRA, 2601.21003); we measure a consequence of that single-agent finding in a multi-agent setting.

1.6. Contributions

1. A 10-domain ordered-pair collaboration grid at Qwen3-1.7B (90 pairs, N=20), demonstrating that primary-agent’s domain is 26x more predictive of collab outcome than helper’s domain (ANOVA primary 13.1%, helper 0.5%).

2. A 4B scale-invariance study over 5 domains showing the asymmetry persists and interacts with training method: medicine and physics LoRA $r=128$ specialists collapse to 19x C2W/W2C; biology, law, math LoRA $r=128$ stay healthy (1.4-2.0x); full FT is healthy on all 5.
3. A rank sweep $r=4..128$ at 1.7B and $r=16..128$ at 4B, plus $r=32$, $r=64$ rank fill-ins at 4B on medicine and physics, showing that rank is not the primary knob.
4. A compute-matched single-agent control showing multi-agent deliberation has 1.4x scaling value beyond compute on base models (+21 pp vs +15 pp).
5. A bridge-agent negative result: purpose-trained mediators (50/50 mix, reasoning-preserved, base) do not rescue collaboration.
6. Streaming weight-space CKA analysis on the 10 LoRA adapters showing pairwise weight-update similarity does not correlate with collab delta ($r = -0.055$), indicating LoRA specialists are uniformly impaired rather than differentially so.

1.7. Organization

Section 2 situates our work in the multi-agent LLM and LoRA literatures. Section 3 describes the experimental design. Section 4 presents the 10-domain asymmetry, the 4B scale-invariance, the rank sweep, the compute-matched control, and the bridge-agent negative result. Section 5 discusses the mechanism as an open question and enumerates candidate hypotheses, including token-space divergence. Section 6 discusses deployment implications. Section 7 lists limitations and spinoff directions.

2. Related Work

(Section stub - will populate from tasks/lit_update_2026_apr.md. Anchors: the multi-agent debate / mixture-of-agents line (Du, Liang, Wang); the collective intelligence line (Becker, Sunstein, Evans, Bratton); the LoRA structural line (Shuttleworth intruder dimensions, CeRA linear ceiling, PERA bilinear, Bayesian-LoRA calibration); post-training for multi-agent (MALT 2412.01928); C2W/W2C prior use (SID 2510.06843).)

3. Experimental Design

(Section stub - specialist training, matched-solo calibration procedure, alternating chain-of-thought protocol, switch classification.)

4. Results

4.1. Primary-agent’s domain dominates collaboration outcome

Figure 1 summarizes the 10x10 ordered-pair grid at 1.7B. Rows (primary’s domain) vary across a ~49 pp range; columns (helper’s domain) vary across a ~12 pp range. Two-way ANOVA on all 1800 outcomes: primary-domain $F(9,1700) = 29.25$, partial eta-squared 0.131, $p < 0.001$. Helper-domain $F(9,1700) = 1.06$, partial eta-squared 0.005, $p = 0.39$. The interaction is not significant ($p = 0.999$). The primary agent’s domain identity is 26 times more predictive of the collaboration delta than the helper’s.

4.2. Universally-harmed and universally-helped primaries

Four domains are systematically harmed by every helper: medicine (-37.8 pp mean), chemistry (-28.3), physics (-24.4), biology (-19.4). Three domains are systematically helped: philosophy (+11.1), law (+8.9), math (+7.8). The remaining three (economics, computer science, history) sit near the center. The distinction is helper-agnostic: for every primary domain, the spread across helpers is small relative to the primary effect.

4.3. 4B scale-invariance and training-method interaction

Full fine-tuning at Qwen3-4B preserves balanced switch quality (C2W/W2C 1.4-2.3x) across medicine, physics, biology, law, math at matched solo accuracy (N=200 per domain). LoRA at r=128 interacts strongly with the primary’s domain: medicine and physics (the universally-harmed domains at 1.7B) collapse to 19x and 19x respectively; biology, law, math stay at 2.0x, 1.43x, 1.71x respectively. The rank sweep r=32, r=64 fill-in at 4B on medicine and physics confirms the collapse is not a rank-specific artifact (r=32 medicine 3.14x; r=64 medicine 19.0x; r=32 physics 19.5x; r=64 physics 10.67x).

Figure. 4B scale-invariance: full-FT preserves balanced switching across domains; LoRA r=128 does not.

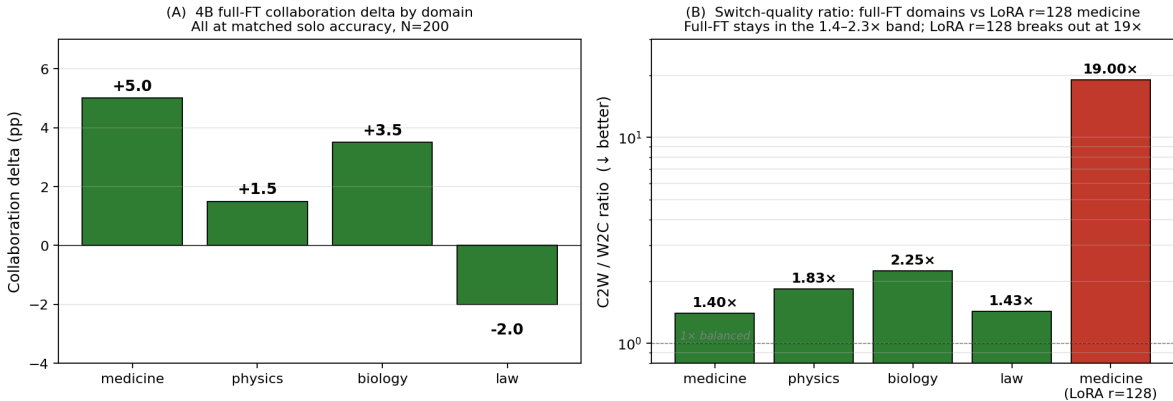


Figure 2: 4B scale-invariance of full-FT collaborativeness across four domains, with LoRA r=128 medicine overlay for contrast. (A) Collaboration delta. (B) C2W/W2C switch-quality ratio.

4.4. Compute-matched deliberation control

Base-model pair deliberation at Qwen3-1.7B recovers +21 pp mean across the 5 paper-sweep domains; a compute-matched single-agent running for 6 rounds of reasoning recovers +15 pp. Multi-agent deliberation has 1.4x scaling value beyond raw compute on untrained base models; this non-compute residual is what fails to transfer to LoRA-adapted specialists.

4.5. Bridge agents fail to rescue collaboration

A naive 50/50-mix LoRA mediator gives -32.8 pp delta with 7.1x C2W/W2C on the medicine+physics pair. A reasoning-preserved LoRA mediator partially repairs (-3 to -3.5 pp, 3.6x). Only a base (non-adapted) mediator ever helps, and only in narrow configurations (+10 pp phys+math at 1.1x C2W/W2C). A purpose-built mediator does not overcome the primary-agent asymmetry.

4.6. Pairwise weight-space similarity does not predict collab delta

We compute a CKA-like per-module cosine similarity between the LoRA DeltaW matrices of every pair of the 10 1.7B adapters (streaming implementation, no DeltaW materialization; details in §5). Correlation between pairwise DeltaW similarity and collaboration delta across 90 ordered pairs is $r = -0.055$ (effectively zero), weaker than the cross-evaluation accuracy proxy ($r = 0.197$). Specialists appear uniformly impaired in weight space rather than differentially so; the variance that matters is at the primary-agent level, not the pair level.

5. Mechanism (open question)

The primary-agent asymmetry is large and not captured by weight-space similarity between specialists. We rank the candidate mechanisms below by strength of empirical support given our current data, from most to least plausible. Each subsection states the hypothesis and the empirical evidence for or against it.

5.1. Primary’s training-token entropy (best partial signal)

Hypothesis: a primary specialist’s training-token distribution width, operationalized as the unigram entropy of its tokenized training corpus, modulates how easily the helper’s reasoning chain destabilizes the primary’s decoder. Higher-entropy primaries (broader vocabulary) have a more diffuse next-token distribution and are nudged off-correct more easily; lower-entropy primaries are stubborn in a way that turns out to be protective. *Empirical support:* across 10 domains, $r(\text{primary training-token entropy, row-mean delta}) = -0.380$ ($n=10$). Medicine (entropy 6.63 nats, broadest vocabulary) is most universally harmed (-37.8 pp); math (5.22 nats, narrow) is universally helped (+7.8 pp). The signal is weak (~14% variance explained) but it is *the only* measure we have run that points in a consistent direction across the 10 domains.

5.2. Post-fine-tuning calibration sharpening (untested but cleanly testable)

Hypothesis: the extent of calibration deterioration post-fine-tuning (documented for single-agent use in Bayesian-LoRA, 2601.21003) is primary-domain-specific. Universally-harmed domains correspond to the largest calibration sharpening; an over-confident primary cannot integrate the helper’s signal because its prior precision dominates the likelihood update. *Empirical support:* not yet measured directly. Adjacent evidence: the 4B LoRA $r=128$ C2W/W2C ratio (19x on medicine, 1.4-2.0x on biology/law/math) is consistent with this hypothesis if calibration sharpening tracks domain identity. *Status:* the cleanest follow-up test we have not yet run; spinoff paper candidate ([paper/spinoff_calibration.md](#)).

5.3. Intruder dimensions (ruled out at the pairwise level)

Hypothesis: the “intruder dimensions” Shuttleworth et al. (2410.21228) document for LoRA-vs-full-FT differ in magnitude or density across domains, and the universally-harmed primaries accumulate more of them. *Empirical support:* the streaming weight-space CKA on the 10 LoRA adapters (§4.6) gives $r = -0.055$ between pairwise weight-space similarity and collab delta. Specialists are uniformly impaired in weight space rather than differentially. *Verdict:* the pairwise version of this hypothesis is ruled out. Per-adapter singular-value spectra could still differ; we have not measured that.

5.4. Bilinear rank constraint (ruled out at the rank levels we tested)

Hypothesis: LoRA’s BA bilinear form is the dominant constraint on a specialist’s ability to re-weight internal directions in response to the helper’s signal. *Empirical support:* our 4B LoRA $r=128$ data shows biology, law, and math specialists collaborating healthily at the same rank where medicine and physics collapse. The bilinear form is identical across all five; only the domain differs. *Verdict:* rank constraint at $r=128$ is not the dominant variable in our data. The role of LoRA’s parameterization at *much* higher rank ($r=512$, $r=1024$) remains an open follow-up question; preliminary $r=256$ / $r=512$ data is in progress.

5.5. Summary of mechanism status

The strongest empirical signal we have is *primary’s training-token entropy* (§5.1, $r=-0.380$), and it explains only $\sim 14\%$ of the variance. *Calibration sharpening* (§5.2) is the cleanest candidate to test next. Both *intruder dimensions* (§5.3) and *bilinear rank constraint* (§5.4) have been weakened by our data. Pinpointing the mechanism is left to follow-up work; the contribution of this paper is the asymmetry phenomenon and the negative results above.

6. Discussion

(Stub - deployment implications, partner-selection heuristics that use the primary-agent’s domain-robustness profile.)

7. Limitations and Future Work

The 10-domain grid at 1.7B is $n=20$ per pair, so individual-pair estimates carry wide confidence intervals (mean CI half-width ~ 18 pp). The aggregate row-level and ANOVA findings are robust to this noise, but individual pair-level claims are not. 7B replication is pending on the same infra. We stop at $r=512$ in the LoRA rank sweep. A fuller mechanism study is left to follow-up work; we stub four independent spinoffs below.

7.1. Future work: post-fine-tuning calibration across domains

(Spinoff - *paper/spinoff_calibration.md*)

7.2. Future work: token-space divergence as mechanism

(In-progress analysis, may become §5 when data lands, otherwise spinoff.)

7.3. Future work: information-theoretic framework

(*paper/spinoff_it_bayesian.md*)

7.4. Future work: training-data composition as a design variable

(*paper/spinoff_training_data.md*)

References

(Bibliography stub. Primary anchors: Shuttleworth et al. 2410.21228; Du et al. 2023; Bayesian-LoRA 2601.21003; SID 2510.06843; MALT 2412.01928; Becker et al. 2017.)